Lecture 3

Topics on NHST

October 24, 2024

2110203 COMP ENG MATH II (CEDT)

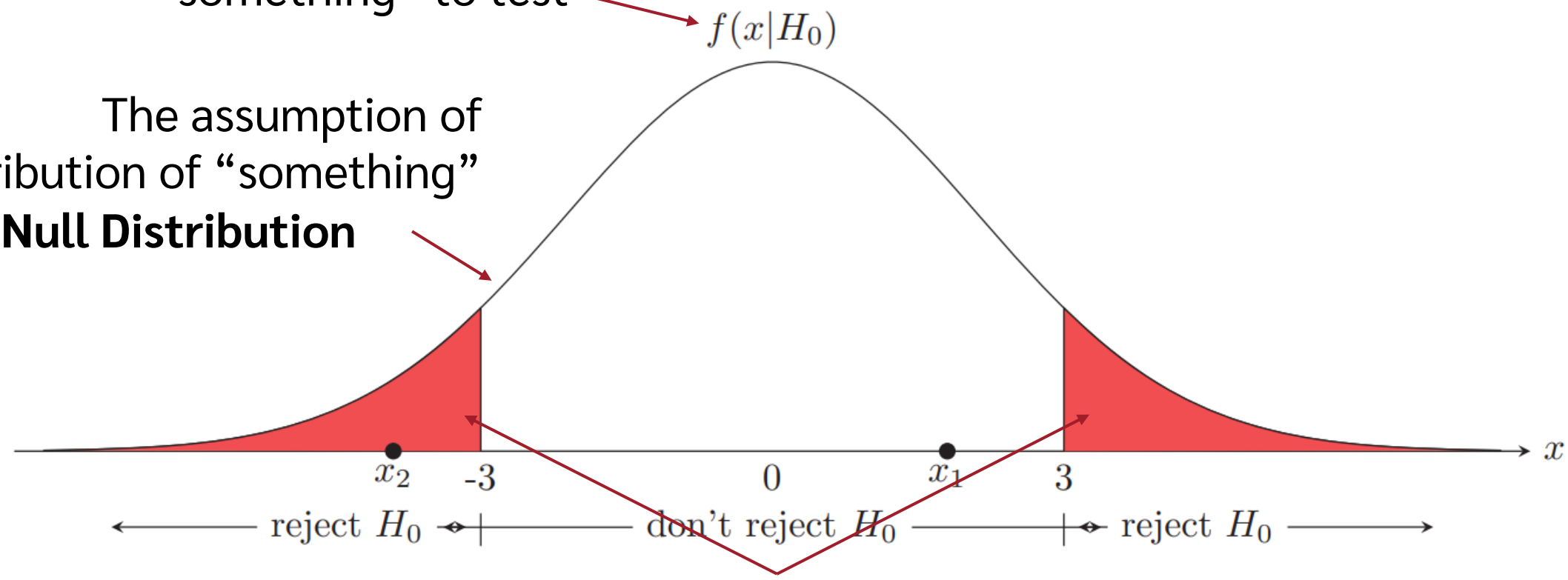
CHULA
COMPUTER
ΣNGINEERING

Recap

Null Hypothesis Statistical Testing

Test Statistic
"something" to test

The assumption of distribution of "something"
Null Distribution



"something" probably doesn't distribute this way

Rejection Region

Null Hypothesis Statistical Testing

P-value: reject H_0 when $p < \alpha$.

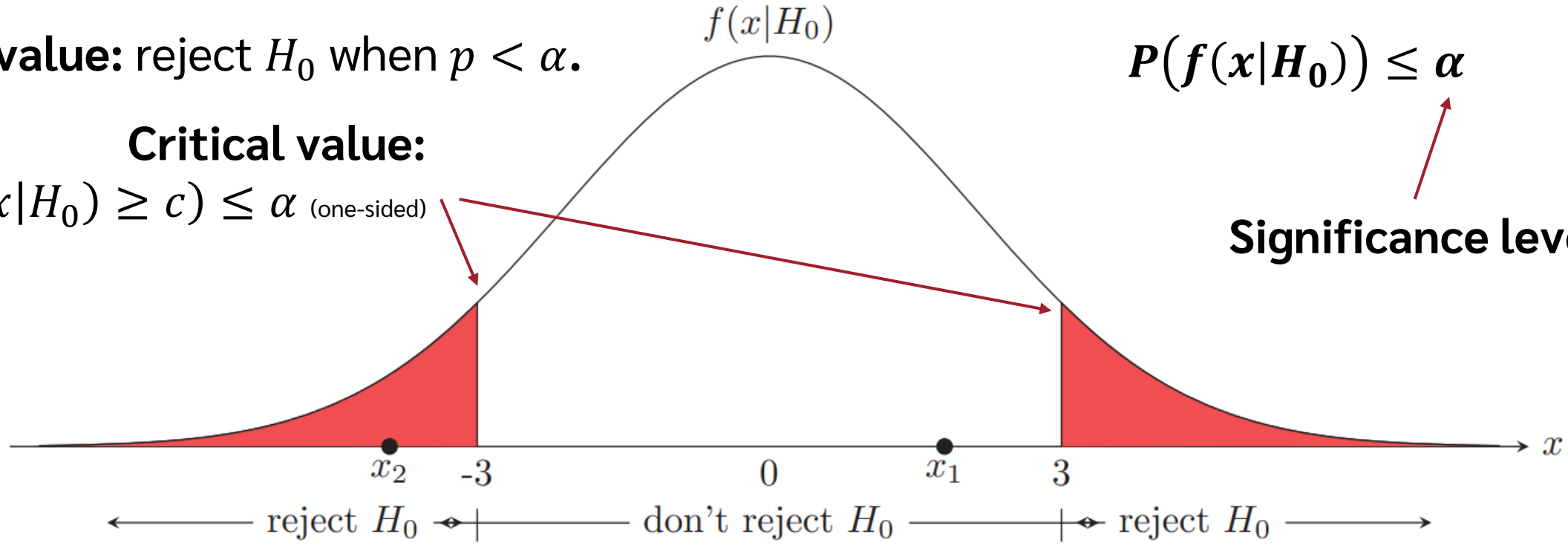
Critical value:

$P(f(x|H_0) \geq c) \leq \alpha$ (one-sided)

Rejection region:

$\boldsymbol{P(f(x|H_0)) \leq \alpha}$

Significance level



Designing a Hypothesis Test

1. Pick a **null hypothesis** H_0 and **alternative hypothesis** H_A.
2. Decide if H_A is **one-sided** or **two-sided**.
3. Pick a **test statistics**.
4. Pick a **significance level** and determine the **rejection region**.
   - Common value are $\alpha = 0.1, 0.05, 0.01$.
5. Determine **the powers**.

Be careful about hypothesis

- When $\alpha < p$: reject H_0
  - Does not mean H_A is true!
- When $\alpha > p$: don't reject H_0
  - Does not mean H_0 is true!

Different type of Statistical Testing

Must-know

- **Z-test**
  - Compare **means against known means**, known equal variance.
- **One-sample t-test**
  - Compare **means against known means**, unknown equal variance.
- **Two-sample t-test**
  - Compare **means of two data sets**, unknown equal variance.
- **Paired t-test**
  - Compare **means of two data sets**, data in pair.

Other type of tests

- **Welch's t-test**
  - Comparing mean with unequal variance
- **F-test for equal means** (One-way ANOVA)
  - Does M sets of (normal) data have equal means (assume same variance)?
- **F-test for equal variance**
  - Does 2 sets of (normal) data have equal variance?
- **χ^2-test (Chi-squared test)**
  - Goodness of fit (does the distribution fits the data?)
  - Homogeneity (does two set of data come from same distribution?)
  - Likelihood-ratio test (which model fits the data better?)
- And more… Just Google.

Example 1

Problem. Suppose the average height of Thai population is 153cm. We measured the average height of this class (280 people) at 156cm. Is this class significantly taller than the population?

What type of test is needed?
(Test what against what, under what condition?)

Example 2

Experiment A: Is the drug effective?

	Dead	Alive	Total
Drug A	9	31	40
No drug	10	30	40

What type of test is needed?
(Test what against what, under what condition?)

Experiment B: Is the drug effective?

	Dead	Alive	Total
Drug A	0	5	5
No drug	10	30	40

Example 3

Problem. Thailand population is 51.5% female. If a random survey of 500 participants covered 246 females and 254 males, is the survey bias?

What type of test is needed?
(Test what against what, under what condition?)

Example 4

Problem. Is there a significant improvement of CEM2 STATS exam score over the STATS COMP ENG exam score?

What type of test is needed?
(Test what against what, under what condition?)

Example 5

Problem. 10 participants were asked to rate 5 drawings from 1 to 5 stars. Does all drawings have the same average rating?

What type of test is needed?
(Test what against what, under what condition?)

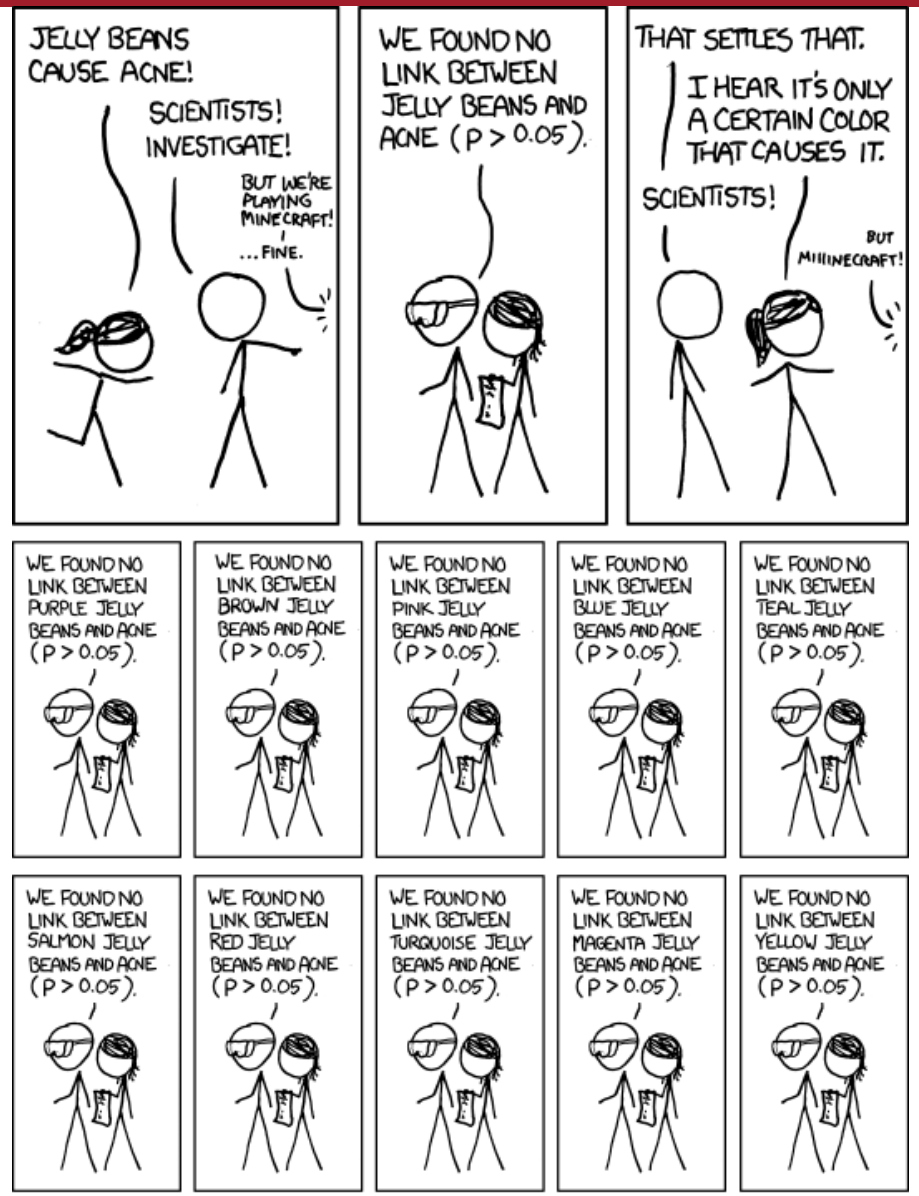
10-MIN BREAK

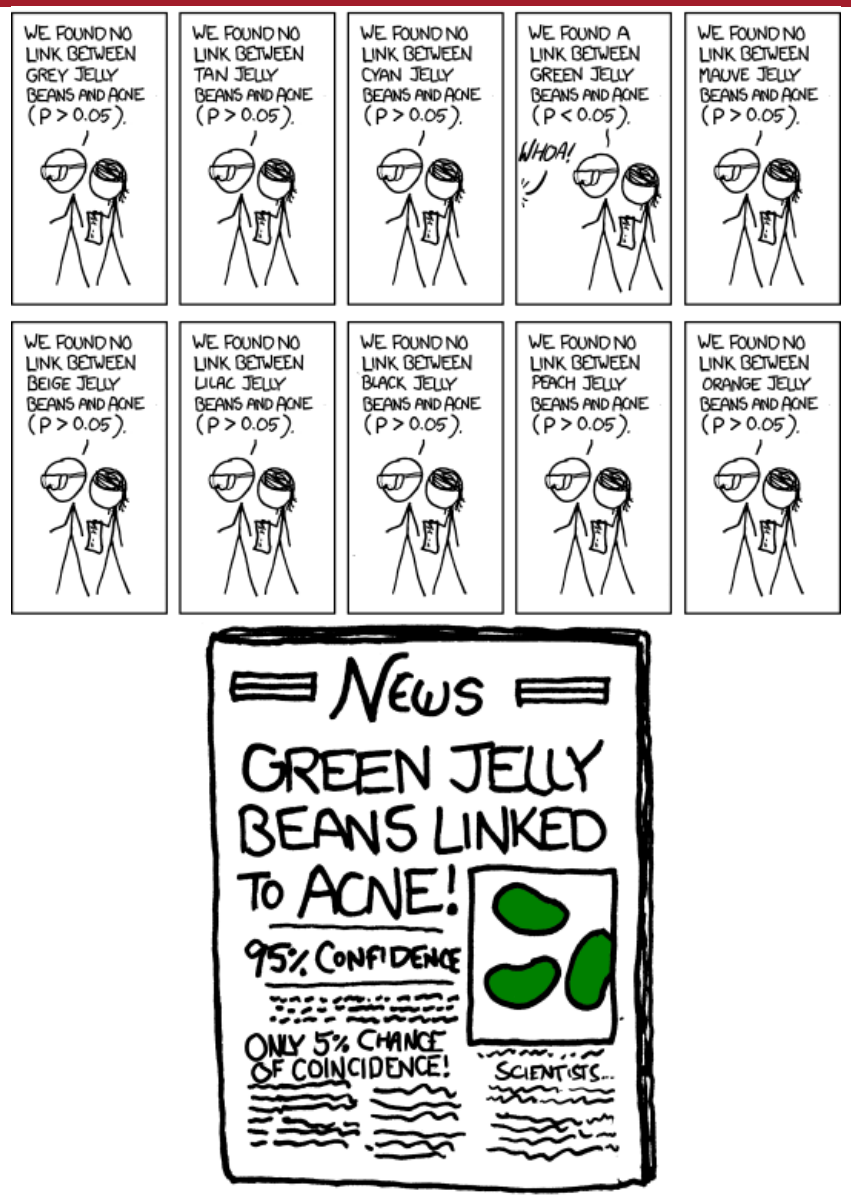
Problem and Best Practice

Problem of p-value

- P-hacking
- Base rate fallacy
- Low power experiments

P-hacking





https://xkcd.com/882/

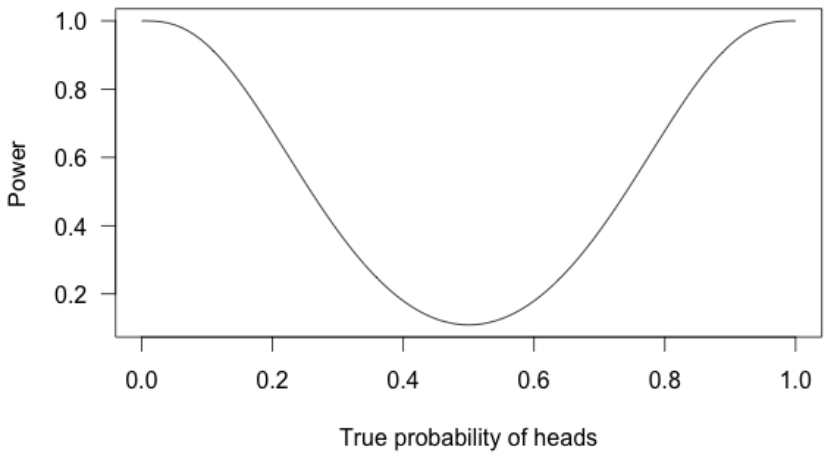
FWER and FDR

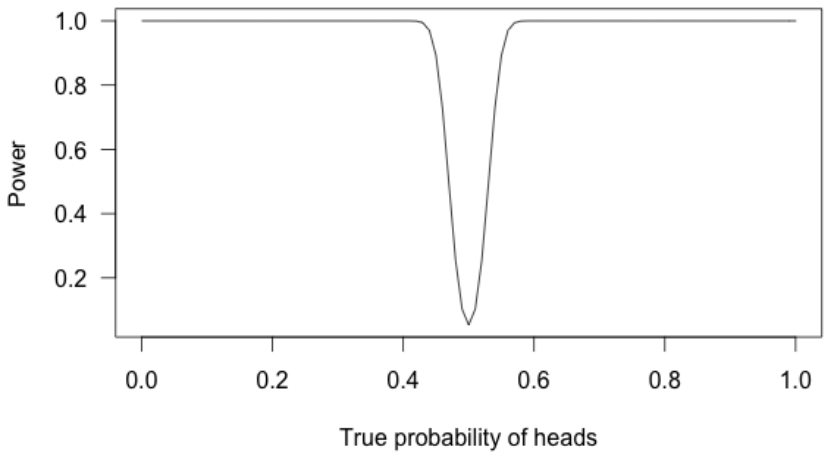
- **Family-wise Error Rate (FWER)**
  - Control $P(No.of\ False\ Positive > 0)$.
  - **Bonferroni Procedure:** $p_i \leq \frac{\alpha}{m}$.

- **False Discovery Rate (FDR)**
  - Control $\mathrm{E}(Q)$ when $Q = \frac{No.of\ False\ Discoveries}{No.of\ Discoveries}$
  - **Benjamini-Hochberg**: $p_i \leq \frac{k\alpha}{m}$ where k is the largest that satisfy $p_k \leq \frac{k\alpha}{m}$

Low Power Experiments

- If the test has low power (underpowered study), no-significant is likely to be due to not enough samples to detect small differences.
- Detected effect often inflated, or biased





https://www.statisticsdonewrong.com/power.html

Misconceptions

- P value tells information about the effect size
- Large P value means no effect

	Treatment 1 (mean ± SD, *n*)	Treatment 2 (mean ± SD, *n*)	Difference between means	*P* value	95 % CI of the difference between means
Experiment A	1,000 ± 100, *n* = 50	990.0 ± 100, *n* = 50	10	0.6	−30 to 50
Experiment B	1,000 ± 100, *n* = 3	950.0 ± 100, *n* = 3	50	0.6	−177 to 277
Experiment C	100 ± 5.0, *n* = 135	102 ± 5.0, *n* = 135	2	0.001	0.8 to 3.2
Experiment D	100 ± 5.0, *n* = 3	135 ± 5.0, *n* = 3	35	0.001	24 to 46

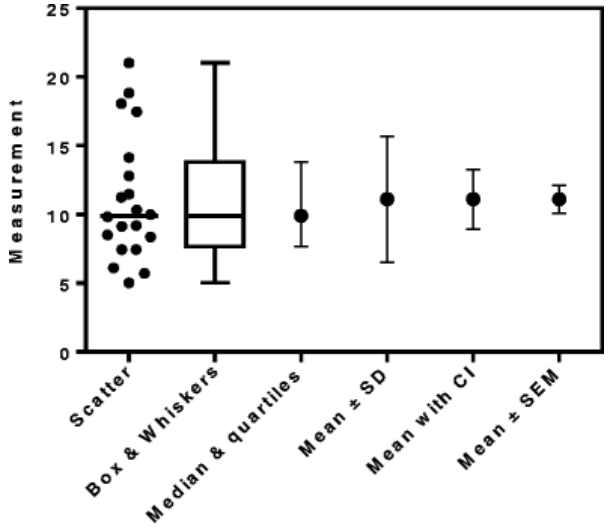
Experiments A and B have identical *P* values, but the scientific conclusion is very different. The interpretation depends upon the scientific context, but in most fields experiment A would be solid negative data proving that there either is no effect or that the effect is tiny. In contrast, experiment B has such a wide confidence interval as to be consistent with nearly any hypothesis. Those data simply do not help answer your scientific question

Similarly, experiments C and D have identical *P* values, but should be interpreted differently. In most experimental contexts, experiment C demonstrates convincingly that while the difference is not zero, it is quite small. Experiment D provides convincing evidence that the effect is large

https://link.springer.com/article/10.1007/s00210-014-1037-6/tables/1

Good Practice

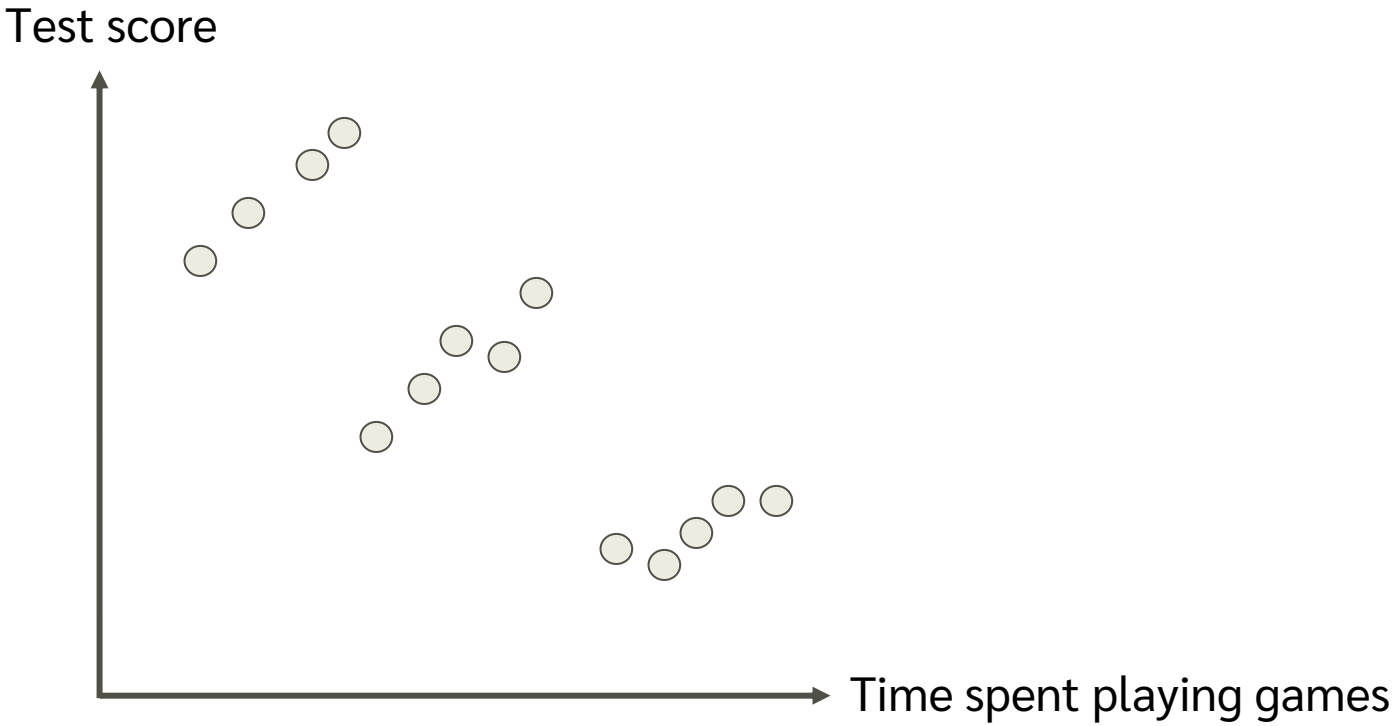
- Always calculate the sample size before completing the experiment
  - Describes the method completely (how do you get the statistics, what test, etc.)
- No need for Hypothesis testing if you do not need to make a **decision** based on the test.
  - Present the story about the data instead, distribution plots, etc.
  - Might use testing on small differences
- Be careful about using the word “significant” in technical writing
- Understand how you present data
  - Presenting the data distribution
  - Presenting the mean estimation



https://link.springer.com/article/10.1007/s00210-014-1037-6/figures/5

Causation vs Correlation

- What can we summarise from this data?



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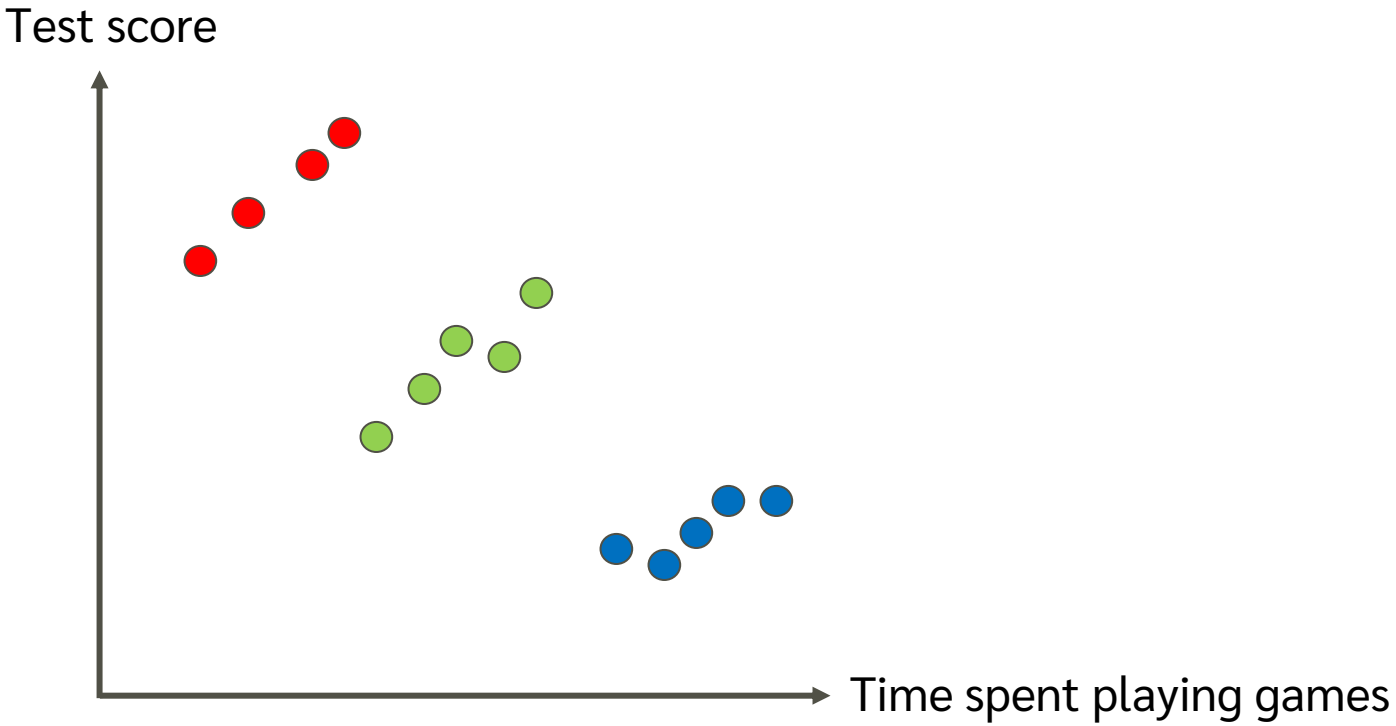
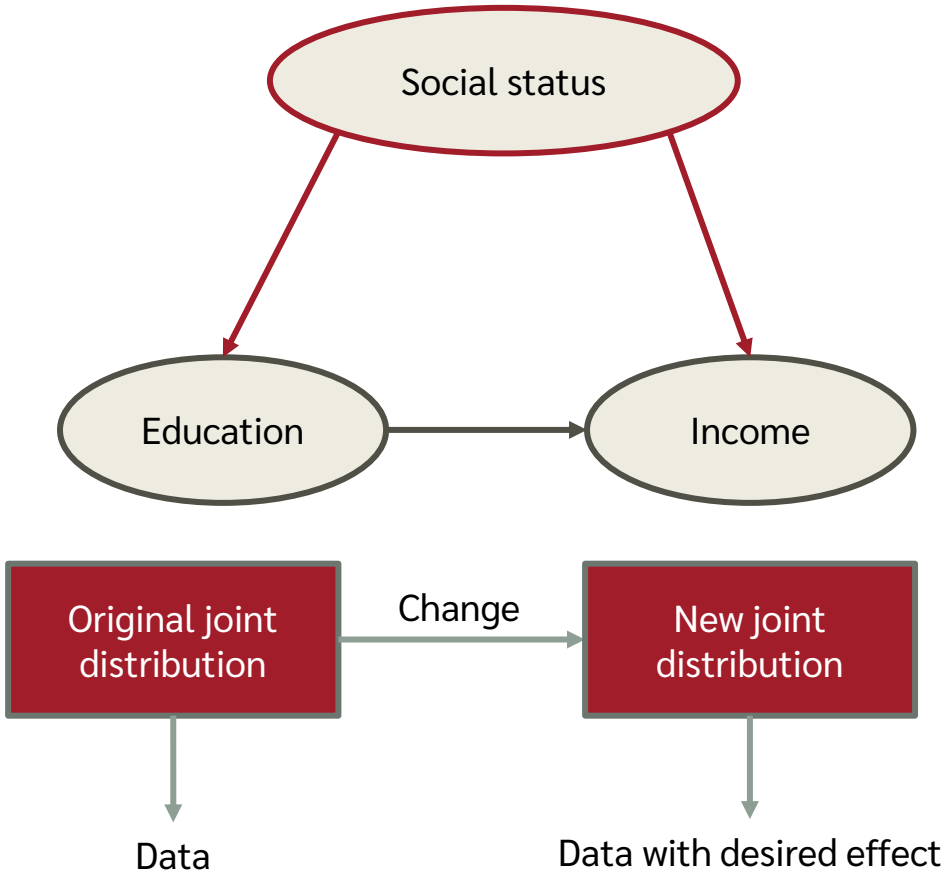


The Challenge of Interventional Inference

- We never really observe the effect of an action



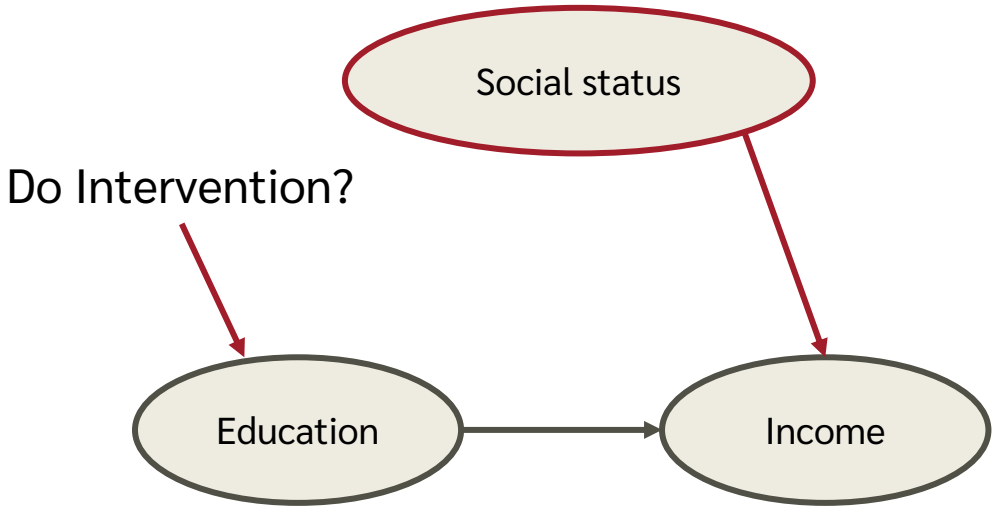
We observe that
High education correlate with High Income

There are hidden factors!

The Challenge of Interventional Inference

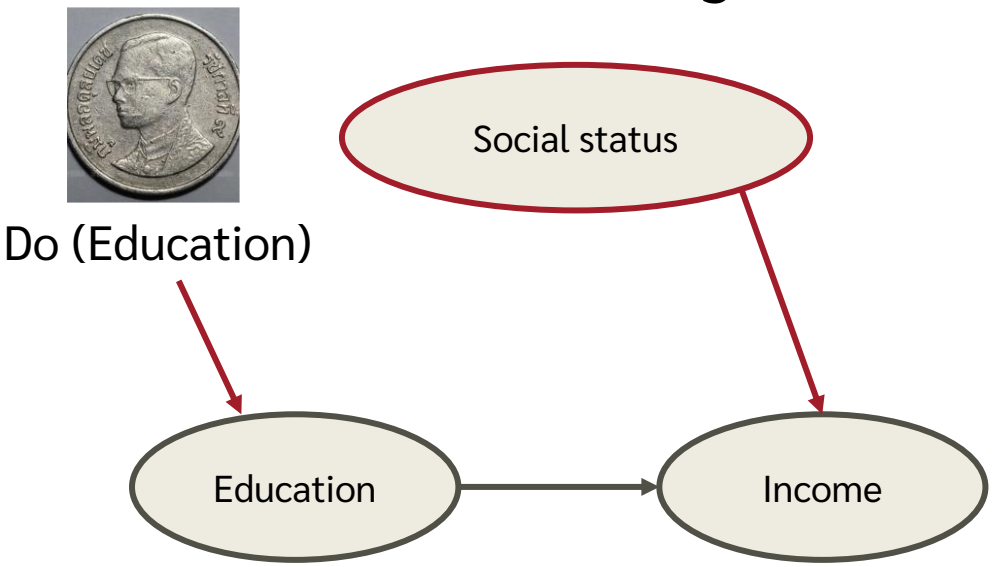
- Disconnect the factors?

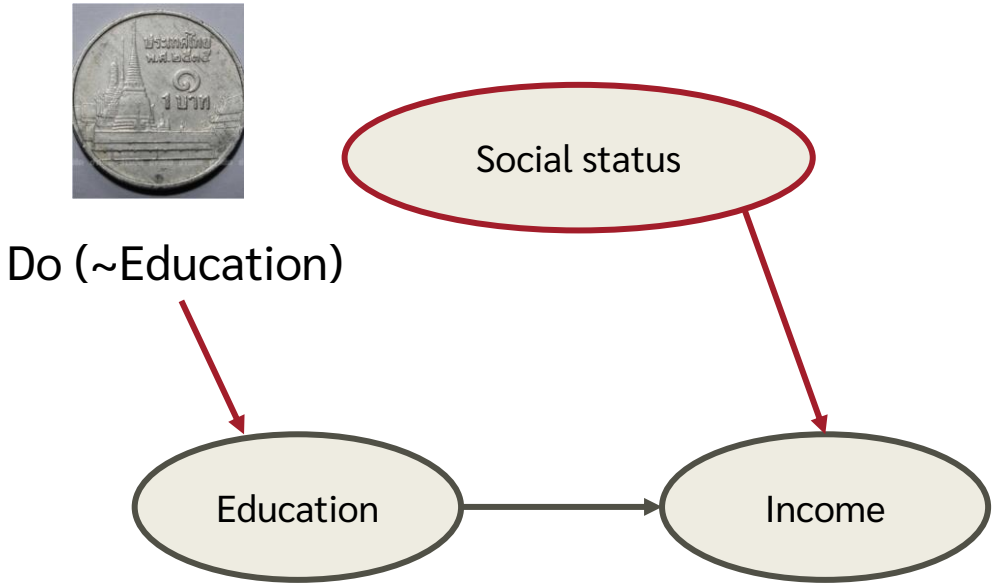
How?



Randomized Control Trial (RCT)

- Disconnect by interacting directly with education.
  - Toss a coin. If **HEADS** they get to go to school.
- Standard for social studies.
  - A form of A/B testing!





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